

Deep Learning Study Tracker

A 20-Week Structured Study Plan

Week	Focus Area	■ Done
Week 1	Python Basics, Linear Algebra, and Calculus Foundations	■
Week 2	Probability, Statistics, and ML Overview	■
Week 3	Supervised vs. Unsupervised Learning, Model Evaluation	■
Week 4	Intro to Neural Networks and Perceptron	■
Week 5	Activation Functions, Backpropagation, and Gradient Descent	■
Week 6	Optimization and Regularization Techniques	■
Week 7	Build Neural Networks using PyTorch or TensorFlow	■
Week 8	Convolutional Neural Networks (CNNs) Fundamentals	■
Week 9	Famous CNN Architectures: LeNet, VGG, ResNet	■
Week 10	Transfer Learning and Image Augmentation	■
Week 11	Recurrent Neural Networks (RNNs) and LSTMs	■
Week 12	Attention Mechanism and Introduction to Transformers	■
Week 13	Working with BERT and Hugging Face Transformers	■
Week 14	Generative Adversarial Networks (GANs)	■
Week 15	Autoencoders and Variational Autoencoders	■
Week 16	Diffusion Models and Advanced Architectures	■
Week 17	Multimodal and Foundation Models	■
Week 18	Model Optimization and Hyperparameter Tuning	■
Week 19	Model Deployment and MLOps Introduction	■
Week 20	Capstone Project: End-to-End Deep Learning Application	■

■ Tips for Success: - Study consistently (2–3 hours daily) - Code every concept you learn - Join ML communities and Kaggle challenges - Read at least one research paper every 2 weeks - Maintain your GitHub portfolio with weekly progress